

INORA

Brain-Computer Interface Project

Team Expansion & Operations Plan

1. Why We Are Expanding

INORA is currently a 5-member team working on word-level EEG decoding — one of the hardest unsolved problems at the intersection of neuroscience, AI, and hardware engineering. To move from research prototype to a functional proof-of-concept, we need more people with specific skills across three domains: Hardware, Software/ML, and Research.

However, we are not expanding just to have more people. We are expanding to build a culture of ownership, initiative, and deep technical thinking. Every new member joins knowing this is not a task-execution role — it is a builder role.

2. The Honest Problem We Are Solving

Before expanding, we acknowledge one truth about our current team:

Team members execute tasks well — but they do not yet propose solutions, suggest new features, or drive their domain independently. This is a structural problem, not a people problem. The system was built for execution, not ownership.

The expansion plan fixes this structurally. Every role, every meeting format, every expectation is redesigned around ownership — not task completion.

3. Target Team Structure (15–20 Members)

We will NOT manage 20 people in a flat structure. That creates chaos. Instead, we use a 3-division layered model:

Level	Role	Manages	Managed By
Founder	Project Lead (You, Annu)	4 Division Heads	Nobody
Layer 1	Hardware Division Head	3–4 Engineers	Founder
Layer 1	ML Division Head	3–4 Engineers	Founder
Layer 1	Frontend/Backend Division Head	3–4 Researchers	Founder
Layer 2	Division Members	Their own tasks	Division Head

The Founder manages only 4 people directly — the division heads. Division heads manage their teams. This keeps decision-making fast and accountability clear.

4. The Three Divisions

Division 1 — Hardware & Signal Acquisition

This team owns everything physical: the EEG circuit, electrode placement, signal quality, and hardware upgrades.

- Refine the EEG amplifier circuit (AD620 and related components)
- Design and build a consistent electrode placement system (swim cap method)
- Improve signal quality — reduce noise, improve contact with electrode gel
- Eventually upgrade from 3-channel to higher channel count
- Document all hardware iterations with photos and schematics

Who should join: Electronics/ECE students, tinkerers, people who have built circuits before.

Division 2 — Software & Machine Learning

This team owns the entire digital pipeline: from raw EEG data to predicted word output.

- Build the trial recording script (stimulus cue → EEG capture → labeled save)
- Build the preprocessing pipeline (bandpass filter, artifact removal, normalization)
- Implement and train the Transformer model on EEG data
- Build real-time inference pipeline
- Maintain the GitHub repository and version control

Who should join: CS students comfortable with Python, ML frameworks (PyTorch/TensorFlow), signal processing.

Division 3 — Frontend & Backend (Web & Interface)

This team owns everything digital that faces the outside world — the INORA website, dashboards, and any web-based interface for visualizing EEG data or project progress.

- Build and maintain the INORA project website
- Design and develop the frontend UI (project showcase, demo pages, team page)
- Build backend systems for data handling, APIs, and any web-based EEG visualization tools
- Integrate real-time or recorded EEG output into a visual dashboard eventually
- Maintain deployment, hosting, and version control for all web assets

Who should join: CS students comfortable with HTML/CSS/JS, React, Node.js, or any backend framework. Designers with coding ability are a bonus.

5. Who We Are Looking For

We recruit for hunger first. Skills second.

Green Flags	Red Flags
Has personal projects outside college curriculum	Asks about certificates or stipend first
Asks about the problem, not the perks	Has never built anything independently
Already curious about neuro/AI/signals on their own	Only joins for resume points
Gets excited about the hard parts of the problem	Wants a defined 9-to-5 task list
Has opinions and proposes things unprompted	Waits to be told what to do

Exception: If someone 1–2 hours away is exceptionally strong and eager — distance becomes flexible. Exceptional beats local, every time.

6. How We Operate — The System

Tool Stack

Tool	Purpose
INORA Web App	Task assignment, progress tracking, sprint management — our internal tool
GitHub	All code — version controlled, reviewed, maintained

Tool	Purpose
Discord / WhatsApp	Daily communication, quick updates
Google Drive	Datasets, research papers, recording files

The Weekly Rhythm

Day	Activity	Duration	Who
Monday	Sprint planning — everyone proposes their week's tasks	30 min	Full team
Wednesday	Quick sync — blockers and progress check	15 min	Division heads + Founder
Friday	Demo — everyone presents what they built this week	30 min	Full team
Sunday	Founder reviews everything privately, plans next sprint	Personal	Founder only

Sprint Structure

Every week has one clear goal. Not ten goals. One.

Example sprint:

- Week Goal: Complete the data recording pipeline
- Hardware: Finalize electrode placement cap and document positions
- Software: Write trial recording Python script with labeled output
- Research: Label 50 trials of word YES and add to dataset
- Founder: Review all outputs + design next sprint problem statements

7. How Ownership Actually Works

This is the most important section. Read it carefully.

We Give Problems — Not Tasks

Wrong way:

'Build a bandpass filter for the EEG pipeline'

Right way:

'Our signal has too much noise. Figure out why and propose a solution by Friday.'

The difference is massive. The first creates an executor. The second creates an engineer.

The Weekly Research Drop

Every member sends one research finding per week to their division head. Not a task completion update — a finding. Something they read, understood, and connected to INORA.

Format: 'I read [paper/article]. Here is what it means for our work. Here is what I think we should try.'

No research drop = no Friday demo attendance. No exceptions.

Meetings Are Theirs — Not Ours

Old format: Founder talks 80%. Team listens. Tasks get assigned.

New format: Team presents 80%. Founder asks hard questions. Team defends their work and proposes next steps.

Nobody says 'I don't know what to do next.' That answer is not acceptable. If you don't know what to do next — that is your problem to solve before Monday.

Public Accountability Dashboard

Every member's weekly deliverable is visible to the entire team on a shared Notion page. When work is private, laziness is easy. When work is public, peer pressure does the job.

8. How Far Ahead the Founder Stays

Layer	What It Means	Example Right Now
Layer 0 — Team	What team is executing this week	Recording script + electrode placement
Layer 1 — Next	What comes after (team doesn't know yet)	Preprocessing pipeline design
Layer 2 — Research	What the Founder is studying now	Which Transformer architecture to use
Layer 3 — Architecture	Full technical vision only Founder holds	End-to-end system design for 50-word decoder

If the team catches up to your thinking — you have lost control of direction. A founder who is only one step ahead gets overruled or confused by their own team. Stay 3 layers ahead. Always.

9. Current Team — Roles and Future Paths

Every current member has been assigned a domain they own completely. Every member has been told explicitly what their future role in INORA looks like. This is not vague potential — it is a named future.

Member	Current Domain	Future Role	They Own
Member 2	Hardware & Signal	Hardware Division Head	EEG circuit, AD620, electrodes, placement
Member 3	Data Pipeline	Software Division Head	Recording script, preprocessing, data management
Member 4	ML Model	ML Lead	Transformer implementation, training, evaluation
Member 5	Research & Docs	Research Division Head	Papers, wiki, dataset labeling, annotation

10. How We Approach New Recruits

We do not send formal recruitment messages. We have a real conversation.

"I'm working on something — decoding words from brain signals using EEG. No fancy equipment, just raw engineering and ML. I saw how you think and I feel you'd find this interesting. Want to hear about it?"

Then explain INORA — the real version. The hard version. The ambitious version. Watch their reaction to the problem, not the startup. If their eyes light up — you have your answer. If they ask 'what's the stipend' first — you have that answer too.

11. What Success Looks Like

At the end of this expansion phase, INORA will have:

- 15–20 members organized across 3 divisions with clear ownership

- A weekly operating rhythm that runs without the Founder initiating everything
- A labeled EEG dataset of 200+ trials across 4–6 words
- A working preprocessing pipeline and initial Transformer model
- A hardware setup with consistent electrode placement and documented signal quality
- A research wiki that any new member can read to get up to speed in 2 days

That is the foundation. From there — we build the 10-word decoder. Then the paper. Then the funding conversation.

INORA — Decoding Human Thought

This document is internal. For team members and recruits only.